SFDCT — Method formulas

A. Block-DCT frequency descriptor

The frequency branch turns a face crop into a compact, localised, normalised descriptor. The crop is divided into non-overlapping 8×8 blocks and the 2-D DCT is applied to each block independently, so a forgery trace stays localised and aligned with the JPEG quantisation grid.

Log-magnitude transform (Eq. 2.1). Each coefficient is rescaled to compress the dynamic range while avoiding the logarithm of zero:

$$D(u,v) = \log\left(1 + \left|F(u,v)\right|\right)$$

The DCT is taken on YCbCr rather than RGB, since most frequency traces sit in the brightness channel Y while the chroma channels C_b, C_r carry complementary compression statistics.

Band assignment (Eq. 2.2). The 64 coefficients of each block are ordered by increasing frequency along a zigzag path and grouped into 16 bands. With the zigzag rank written as rank, a coefficient is assigned to a band by:

$$\mathrm{band}(\mathrm{rank}) = \lfloor \mathrm{rank}/4 \rfloor$$

so four consecutive ranks fall into each band.

Per-band feature (Eq. 2.3). The feature for band b in channel c is the mean magnitude of its coefficients:

$$g_{b,c} = \frac{1}{\left|B_b\right|} \sum_{(u,v) \in B_b} \left|F_c(u,v)\right|$$

where B_b is the set of coefficient positions in band b. Stacking 16 bands across the three YCbCr channels yields a 48-dimensional descriptor ($16 \times 3 = 48$).

B. Gated cross-attention fusion

The two streams are merged by cross-attention in which the spatial features F_s form the query and the frequency descriptor D forms the key and value. Each spatial position queries which frequency traces are relevant in its region and retrieves a weighted summary, so the fusion is selective in space rather than a flat concatenation.

Projections. The query, key, and value are linear projections:

$$Q = W_q F_s, K = W_k D, V = W_v D$$

Attended context (Eq. 2.4). The attended context is:

$$\text{context} = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V$$

where d_k is the key dimension and the $\sqrt{d_k}$ factor keeps the softmax weights from saturating.

Gated residual (Eq. 2.5). The context enters the spatial stream through a gated residual connection whose scalar gate α is initialised to zero:

$$F_{\text{fused}} = F_s + \alpha \cdot \text{context}, \alpha \text{ initialised to } 0$$

At the first step $\alpha = 0$, so $F_{\text{fused}} = F_s$ and the model is exactly the backbone, which realises the floor guarantee of the zero-initialised gate. The gate then behaves as a learned strength control: if the frequency context reduces the training loss the gradient opens the gate, and if it is noisy or useless α is driven back towards zero. The learned value of α is therefore also a direct measurement of how much the frequency branch contributes.